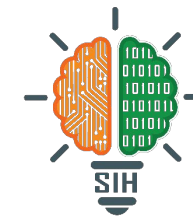
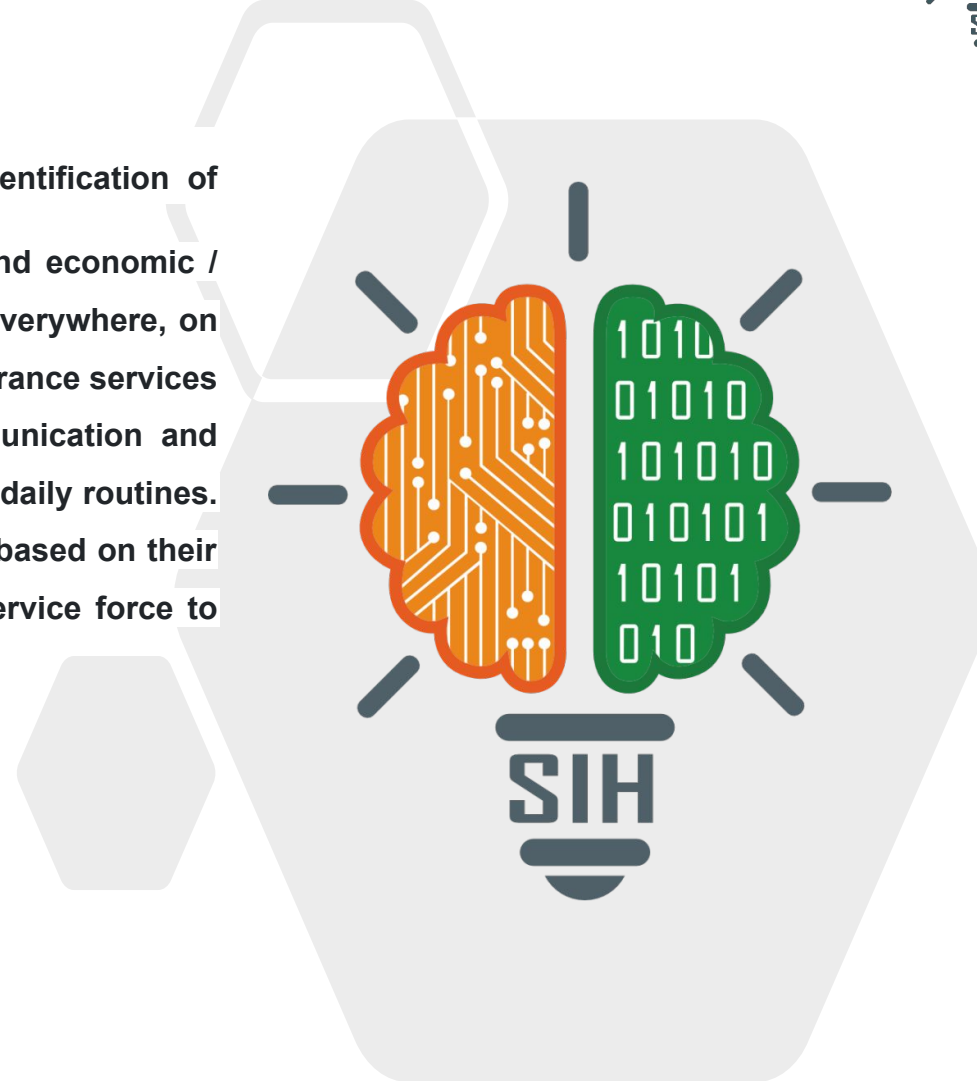


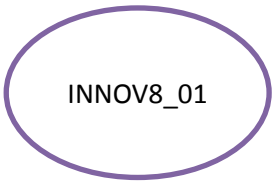
SMART INDIA HACKATHON 2024



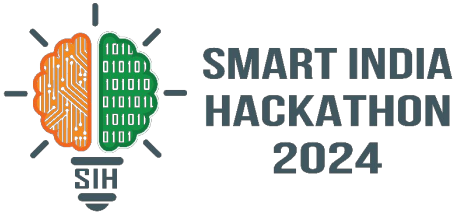
SMART INDIA
HACKATHON
2024

- **Problem Statement ID- 1760**
- **Problem Statement Title-** AI based identification of Financial (Banking & Insurance) needs based on demography and economic / farming Cycle "While the Post Office Network serves everyone, everywhere, on all days, the need for various services including financial and insurance services has a seasonal variation to it. There is a need to time communication and support to customers at times they require it most based on their daily routines. A robust system to segment, target and focus on each customer based on their needs from time to time is needed for India Post to direct its service force to make efforts in response to such needs.
- **Theme- Miscellaneous**
- **PS Category- Software**
- **Team ID- 7505**
- **Team Name- INNOV8_01**





डाक Setu: Precision Financial Services via AI and Postal Networks



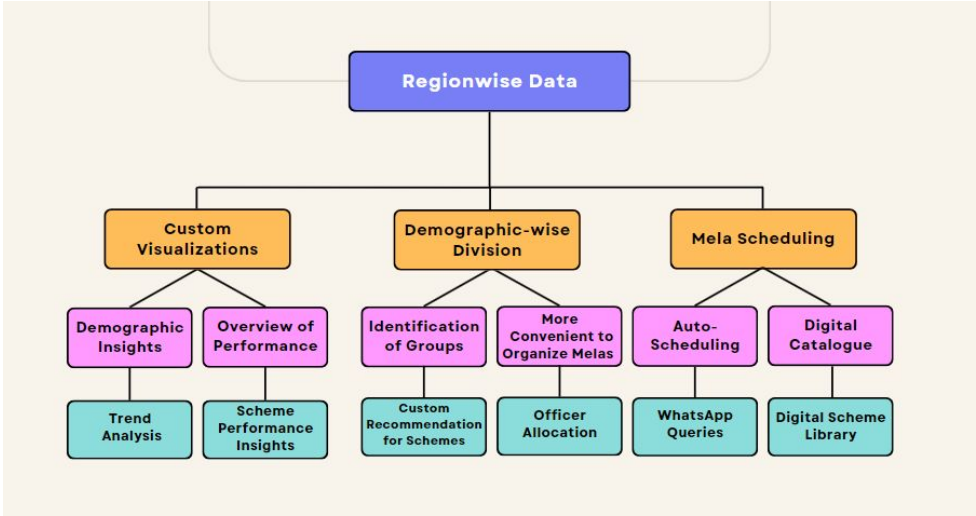
Idea/Approach:

DaakSetu is a digital platform designed to enhance Post Office Savings Bank outreach by leveraging demographic insights and real-time data. It helps post offices plan targeted melas, track performance, and optimize workforce management for maximum impact.

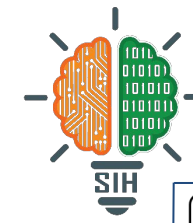
Feature	Description
WhatsApp Bot for Queries	24/7 chatbot for assistance on POSB schemes, mela planning, and automated recommendations.
Digital Cataloging with Filtering	Digital library of schemes with filtering by demographics and region.
Custom Mela Recommendations	Tailored mela suggestions based on demographics and seasonal trends.
Real-time Mela Progress Tracking	Live dashboard for tracking accounts opened, subscriptions, and performance.
Comparative Post Office Performance	Comparison tool with a leaderboard showcasing top-performing post offices.
Region-wise Demographic Visualizations	Heatmaps and overlays showing key demographics and best-suited POSB products for regions.
Dynamic Work Allocation	Automated task assignment with real-time progress tracking for efficient campaign execution.
AI-based Scheme Recommendations	AI-driven suggestions for schemes based on demographic and regional data for targeted outreach.

Innovation and Uniqueness of Solution:

- 1.Demographic Targeting:** DaakSetu uses real-time data to recommend region-specific POSB schemes, tailoring campaigns to local needs for better results.
- 2.Mela Planning & Tracking:** It offers custom mela recommendations and real-time progress tracking, allowing for on-the-go adjustments.
- 3.Workforce Automation:** Dynamic task allocation and WhatsApp integration streamline staff management, boosting efficiency and responsiveness.



TECHNICAL APPROACH

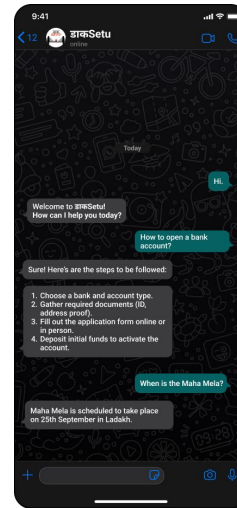
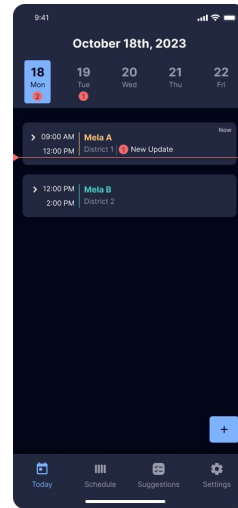
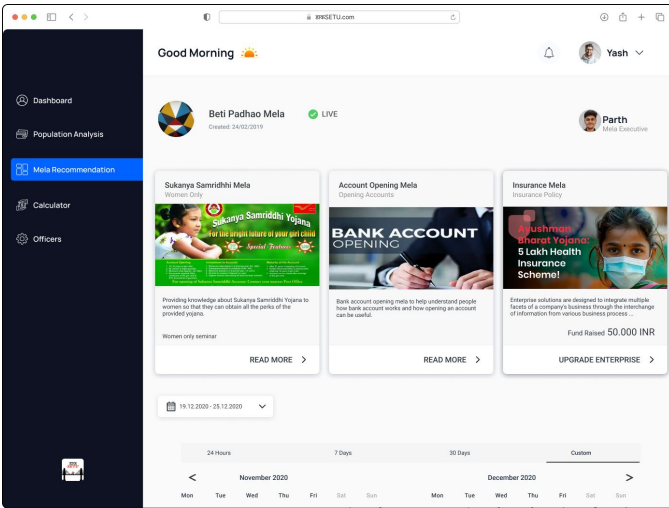


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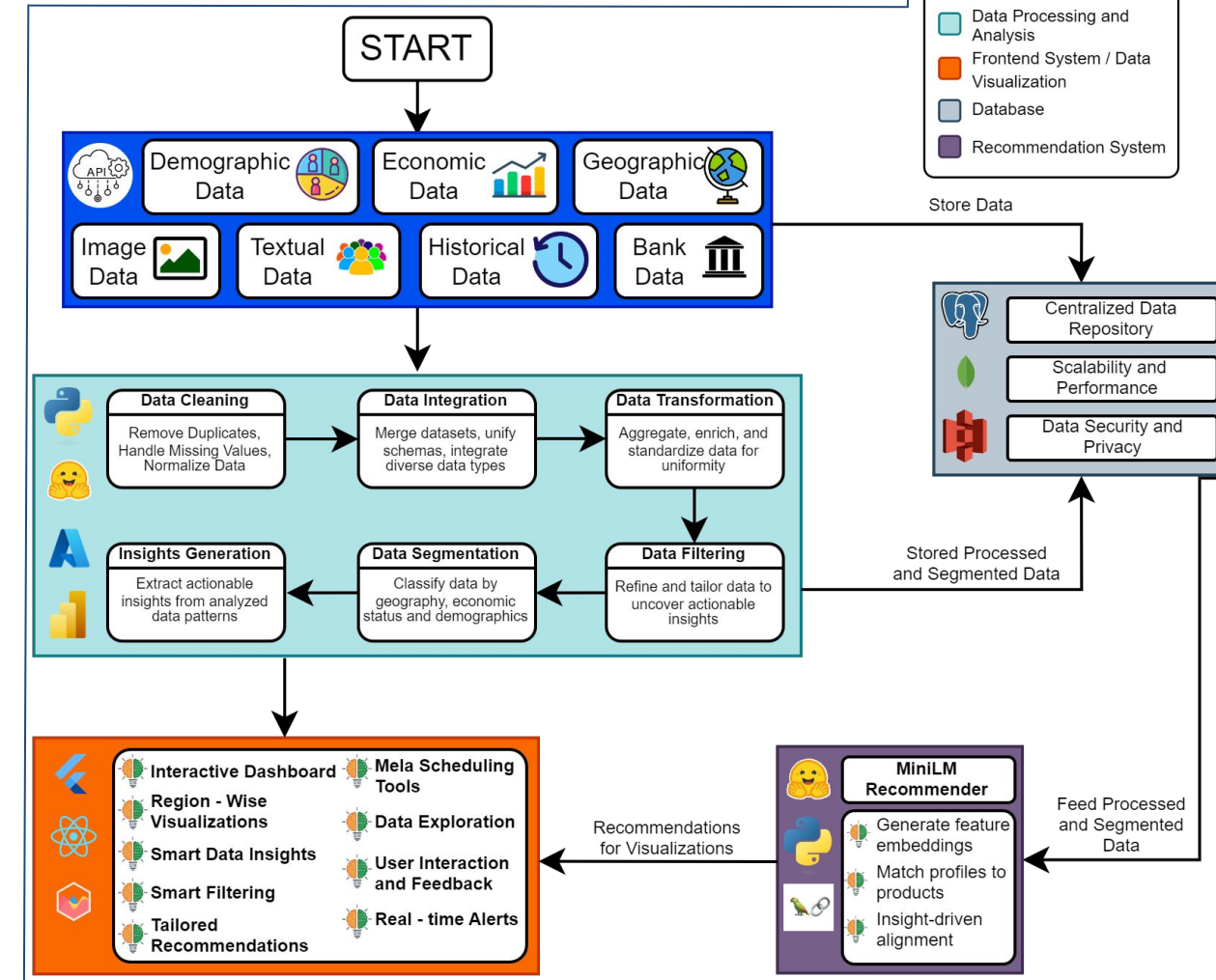
Technology Stack:

- Hugging Face 🤖
- Microsoft Azure 🌐
- AWS 🏠
- Flutter 🦋
- LangChain 🦜🔗
- Flask 🐍
- MongoDB 🌿
- React JS ⚛️
- Python 🐍
- PostgreSQL 🐉
- Node JS 🟩
- Chart.js 📊

Working Prototype:



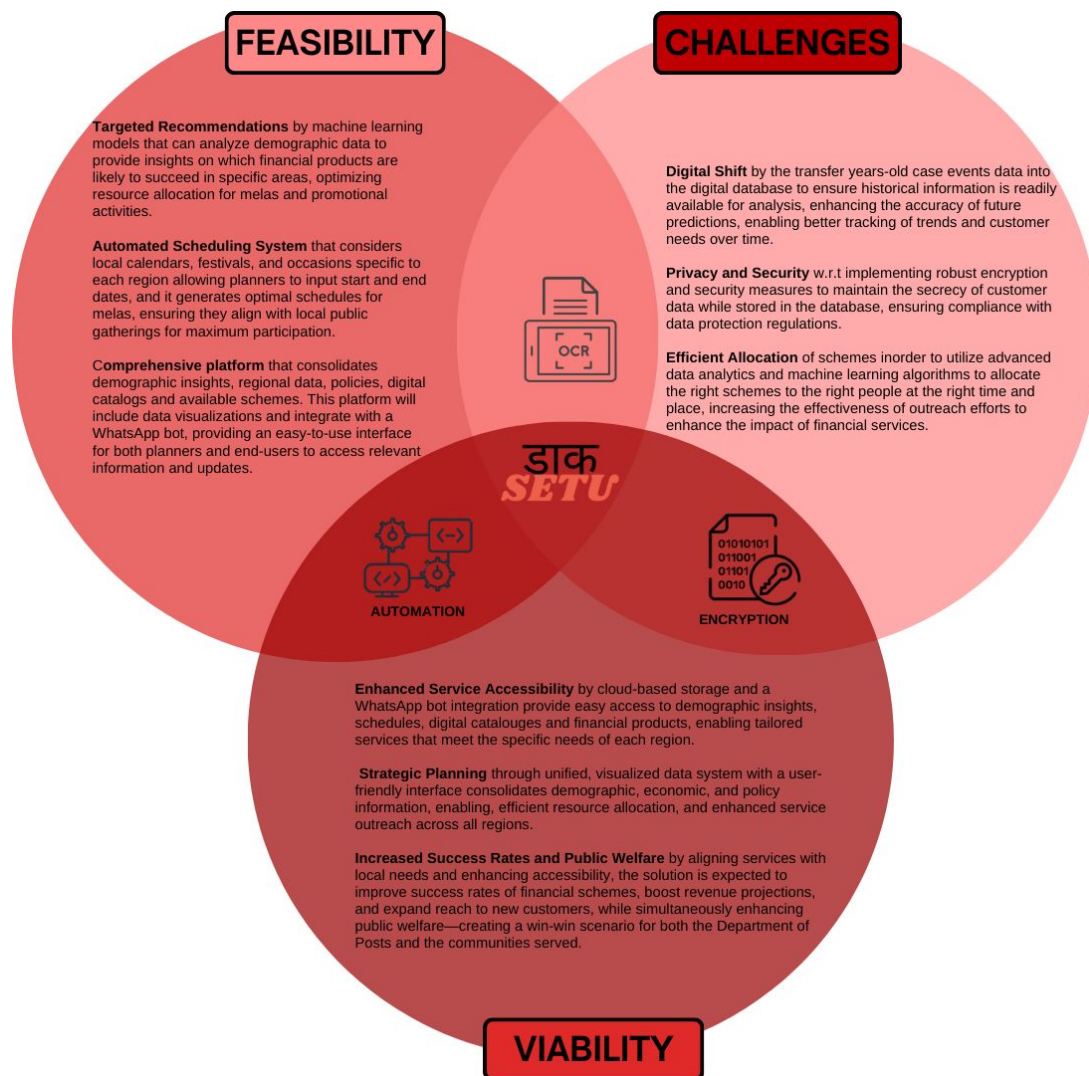
System Workflow / Data Flow Diagram:



Link for the figma design: [डाक Setu](#)

Link for System Workflow/ Data Workflow Diagram: [डाक Setu System WorkFlow](#)

FEASIBILITY AND VIABILITY



Models Explored for Delivering Targeted Solution Functionalities:

1. facebook/bart-large-mnli (For demographic analysis)

Categorizes text descriptions into demographic groups using zero-shot classification, enabling flexible text categorization without the need for labeled data.

2. microsoft/BioGPT (For geographical analysis)

Extracts and categorizes location-based information from unstructured text, making it useful for geo-targeted insights and context-specific data retrieval in large documents.

3. distilbert-base-multilingual-cased (For multi-linguistics)

Performs multi-lingual text classification and sentiment analysis on local feedback, supporting language diversity and enabling feedback analysis in multiple languages.

4. sentence-transformers/all-MiniLM-L6-v2 (For recommendations)

Creates embeddings for personalized recommendations based on scheme features and user profiles, aiding in providing targeted solutions and improving user engagement.

5. google/vit-base-patch16-224 (For pictorial data)

Analyzes and categorizes images from melas and local areas for visual reporting, supporting image-based insights and analysis for event-driven data.

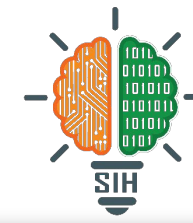
6. distilbert/distilroberta-base (For economic analysis)

A lightweight variant of RoBERTa used for analyzing economic text data. It helps process large volumes of economic reports, providing insights into market trends and forecasts through classification and summarization.

7. microsoft/deberta-v3-large (For time series analysis)

Adapts DeBERTa's transformer-based model for time series text data, allowing the extraction of temporal patterns from textual data and providing better accuracy in time-sensitive tasks like forecasting and anomaly detection.

IMPACT AND BENEFITS



**SMART INDIA
HACKATHON
2024**

Government Benefits

40%

Reduce decision-making time for new scheme launches

25%

Uptick in Kisan Vikas Patra Investments

15%

Rise in Senior Citizens with Savings accounts

20%

Increase in Sukanya Samridhi accounts

15-20%

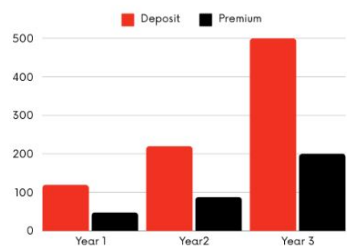
Bump up in customer satisfaction scores

2 million

New mobile banking activations

5500+

Jobs for rural banking positions and data analyst



Potential ₹500 crore boost in deposits and ₹200 crore increase in annual premium collection over 3 years

Environmental and Economical Impact

30%

Reduction in Mela Cost

40%

Reduced Paper Use

BENEFITS OF THE SOLUTION

1. Social Impact

- Potential to reduce economic disparities by providing appropriate financial tools
- Supports government initiatives for social security and women's empowerment



2. Economic Benefits

- Potential for higher conversion rates and reduced customer acquisition costs
- Long-term economic stability for customers through better financial planning



3. Operational Impact

- Optimized resource allocation for marketing events and melas
- Improved performance metrics for individual post offices and regions



4. Technical Impact

- Modernization of India Post's data infrastructure and analytics capabilities
- Potential for integration with other digital initiatives (e.g., Digital India)



5. Business Impact

- Increased market share in rural and semi-urban financial services
- Improved competitiveness against private sector banks and insurance providers

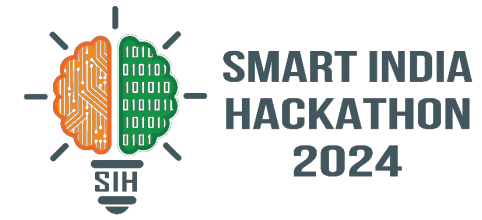


6. Environmental Impact

- Reduced paper waste from untargeted marketing materials.
- Promoting digital financial services, reducing need for physical transportation



RESEARCH AND REFERENCES



Past Melas:

1. <https://www.postalstudy.in/2023/02/posb-maha-mela-from-20th-to-24th.html#gsc.tab=0> - POSB Maha Mela from 20th to 24th February 2023.
2. https://www.uniindia.com/~post-office-savings-bank-maha-mela-in-kerala/States/news/2919983.html#google_vignette - POSB Maha Mela in Kerala

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- [1] <https://www.shiksha.com/online-courses/articles/post-office-saving-schemes-blogId-152093> - a blog on explaining the top 10 Post Office Saving Schemes, listing their key features and highlighting the importance of secured investment.
- [2] https://www.academia.edu/36435267/PERFORMANCE_OF_INDIAN_POST_OFFICE_SAVING_SCHEMES_IN_RECENT_TRENDS - K. Baby saranya, Dr. R. Hamsalakshmi. (2018). The present study gives valuable insight into the investment behavior and preference of Indian investors, particularly towards post office savings schemes. This will definitely help policy framers and regulators while framing policies and schemes to cater to the needs of different investor segments. *Int. J. Adv. Res.* 6(3), 998-1004.
- [3] <https://www.ijrti.org/papers/IJRTI2303007.pdf> - Dr Jitendra Singh Bidawat. (2023). Covers the different programs that the Government of India developed for rural development and discusses how those programs affected rural community life.. This study further contains a table that has provided the outlays according to each scheme and its associated year, with B.E and R.E.. 2023 IJRTI | Volume 8, Issue 3 | ISSN: 2456-3315
- [4] <https://www.ijraset.com/best-journal/a-study-on-consumer-satisfaction-towards-indian-post-office> - Vimal V. (2023). focuses on analyzing the consumer satisfaction towards the Indian Post Office. It provides an overview of the Indian postal system, its history, services, and various postal schemes. ISSN: 2321-9653; IC Value: 45.98; SJ Impact Factor: 7.538 Volume 11 Issue V May 2023.
- [5] <https://www.jetir.org/papers/JETIR2108457.pdf> - Dr.K.BABY SARANYA. (2021). micro-level analysis based on data from Coimbatore region, The study found that low and middle-income families in Coimbatore prefer to save money in post office savings banks, especially for their children's future educational or marriage expenses. They mainly invest in recurring deposit schemes with monthly savings of ₹200 to ₹400. *JETIR August 2021, Volume 8, Issue 8* | (ISSN-2349-5162)